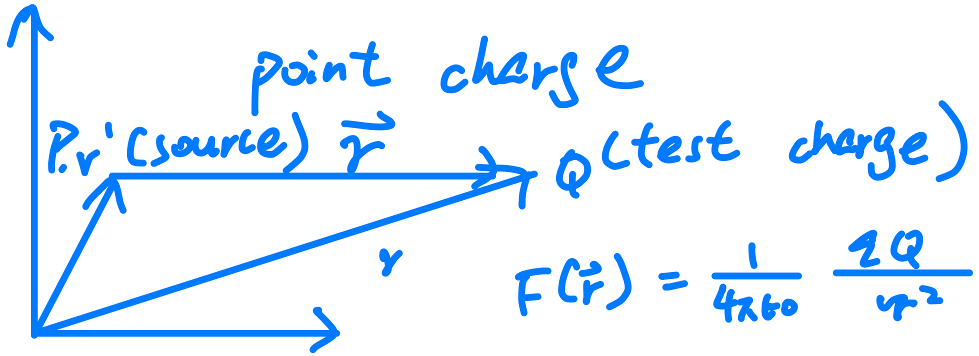
Electrostatics (time independent)

1. Columb's law



$F(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{r}$

ϵ_0 = vacumn permitivety
$8.85 \times 10^{-12} \left(\frac{C^2}{Nm^2}\right)$

[C], a big unit
rub 10^{-6}C or $1\mu C$

Battery : 10^4 C
Quantized : 1.6×10^{-19} C
πC is not possible!
$\Rightarrow$ treat as continuous

- whole universe is electrocially neutral
- total charge is conserved

like energy conservation

- time translation symmetry

momentum conservation

- space symmetry

Charge conversation

- gauge symmetry

- Treats charge as continuous distributed

?. why the for force need to $\propto \frac{1}{r^2}$

- Test result
- Maxwell equations. if $f \propto \frac{1}{r^{2.001}}$, photons will have rest mass!

$$\vec{F}_{tot} = \frac{1}{4\pi\epsilon_0} Q \sum_{i=1}^{N} \frac{q_i}{r_i^2}\hat{r}_i$$

test charge

$$= \underbrace{\left(\frac{1}{4\pi\epsilon_0} \sum_{i=1}^{N} \frac{q_i}{r_i^2}\hat{r}_i\right)} Q$$

$$= E(\vec{r})\, Q$$

Electric field at point $\vec{r}$

$$E(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r^2}\hat{r}$$

☆ $$= \frac{1}{4\pi\epsilon_0}\iiint_V \frac{\hat{r}}{r^2}\rho(\vec{r}')\,dV'$$

2d electrons distributed through
a surface,

$$\rho(\vec{r}') = \delta(\vec{r}')$$

2d electrons distributed through
(1D line

$$\rho(\vec{r}') = \delta^2(\vec{r}')$$

2d point charge

$$\rho(\vec{r}') = \delta^3(\vec{r}')$$

Surface distributed charges $= \frac{1}{4\pi\epsilon_0} \iint_S \frac{\hat{r}}{r^2} \sigma(\vec{r})\, da'$

not vector

- integrate over a surface, as the charges distribute over that!

line distributed charges $= \frac{1}{4\pi\epsilon_0} \int_P \frac{\hat{r}}{r^2} \lambda(\vec{r})\, dl$

- Why linearly holds?

Some counter examples?

Inside blackholes

2. Gauss's law

$$\nabla \cdot \vec{E}(\vec{r}) = \frac{\rho(\vec{r})}{\varepsilon_0}$$

$$\oiint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\varepsilon_0}$$

only this related to x, y, z!

$$\nabla \cdot \vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \iiint_V \left(\nabla \cdot \boxed{\frac{\hat{\mathscr{r}}}{\mathscr{r}^2}} \right) \rho(\vec{r}')\, dV'$$

with superposition principle!

$$= \frac{1}{\cancel{4\pi}\varepsilon_0} \iiint_V \cancel{4\pi}\, \delta^3(\vec{r} - \vec{r}')\, \rho(\vec{r}')\, dV'$$

sifting property!

$$= \frac{\rho(\vec{r})}{\varepsilon_0}$$

exact location of electric divergence

$$\iiint_V \nabla \cdot \vec{E}\, dV = \iiint_V \frac{\rho(\vec{r})}{\varepsilon_0}\, dV$$

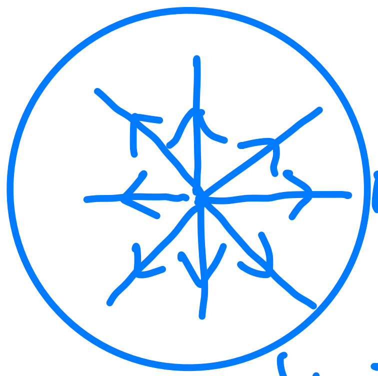
$$\oiint_S E(\vec{r})\, d\vec{a} = \frac{Q_{enc}}{\varepsilon_0}$$

Gauss's law integral form

total flux!

⚠ Cannot add field lines

Only can add field vectors!



density of line ∝ strength

$E(\vec{r}) \propto \frac{1}{r^2}$

↳ Total number of lines are conserved

field line only generated by charge

- continuous

$$\vec{E}_q(\vec{r}) = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}$$

$$= \frac{1}{4\pi\epsilon_0} q \left(\nabla\left(-\frac{1}{r}\right)\right)$$

$$= \nabla\left(-\frac{q}{4\pi\epsilon_0 r}\right)$$

gradient for a scaler function!

one single charge

- multiple charge

$$= \nabla\left(-\frac{1}{4\pi\epsilon_0}\sum_i \frac{q_i}{r_i}\right)$$

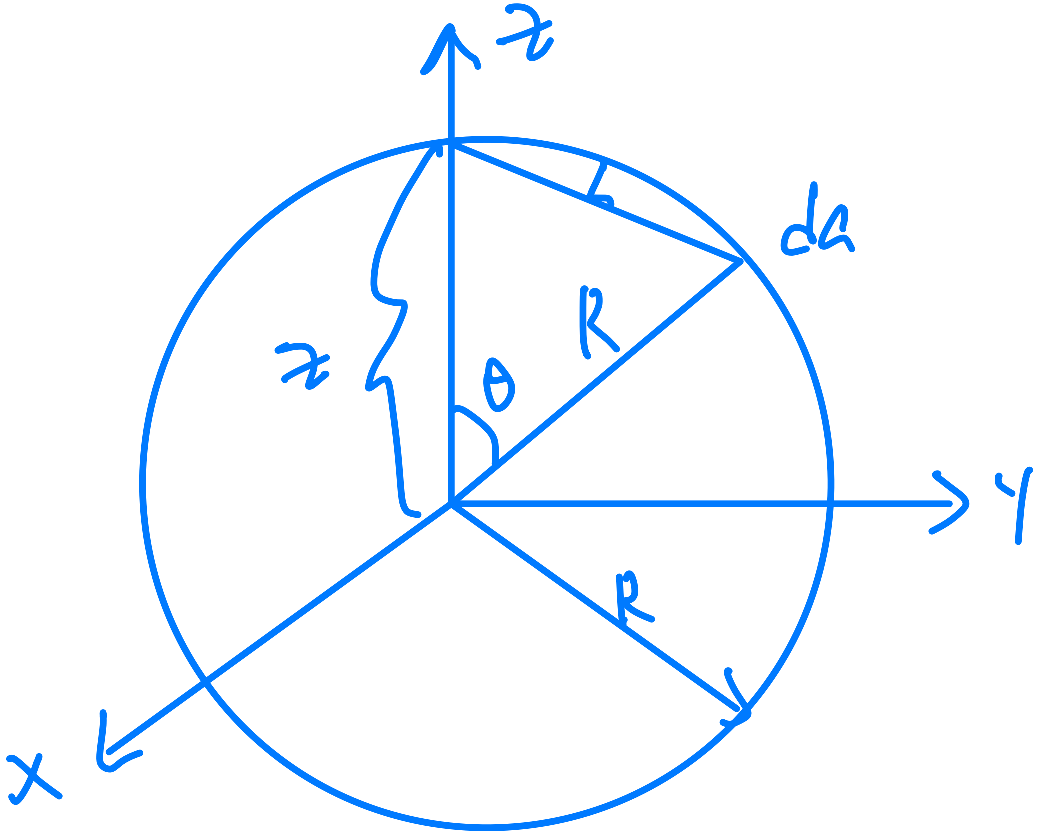
$\vec{E} = -\nabla\phi$, ϕ is potential! Scaler integration

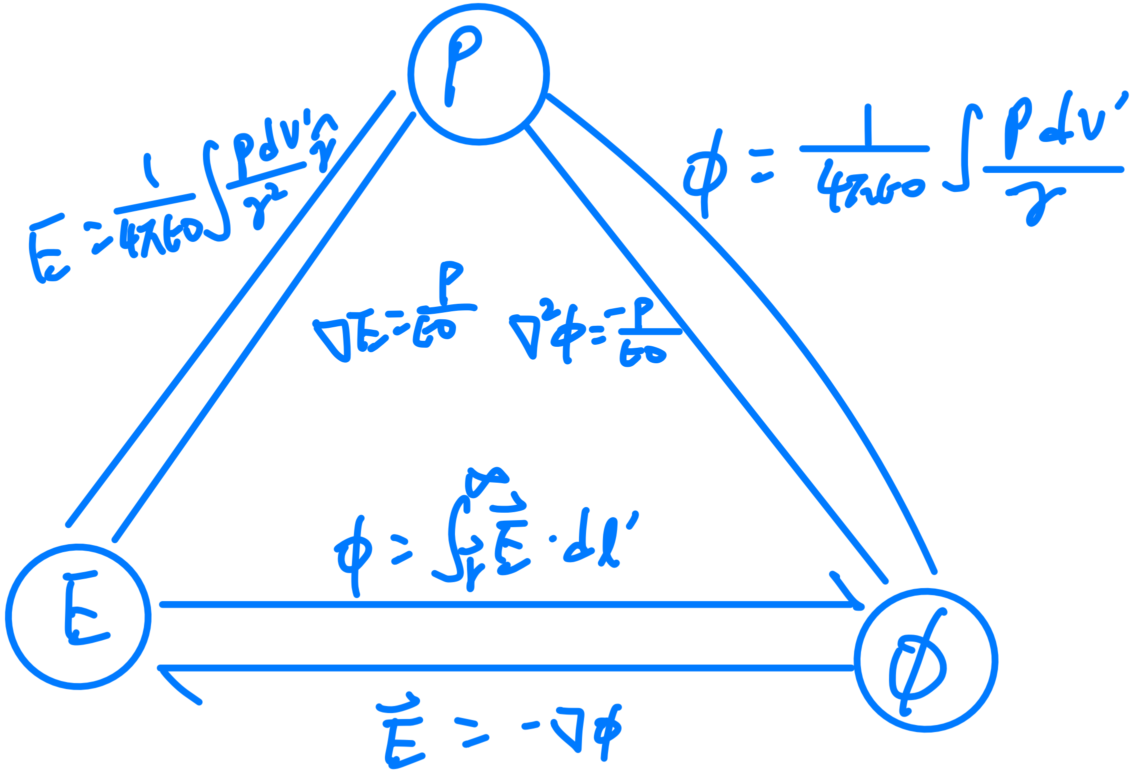
$\phi = \frac{1}{4\pi\epsilon_0}\iiint \frac{\rho(\vec{r}')}{\hat{r}} dV'$ volume distribution $\Rightarrow$ much easier

$\phi = \frac{1}{4\pi\epsilon_0}\iint \frac{\sigma(\vec{r}')}{\hat{r}} da'$ surface distribution

$\phi = \frac{1}{4\pi\epsilon_0}\int \frac{\lambda(\vec{r}')}{\hat{r}} dl$ line distribution

Good news: ϕ is scaler function





$$\vec{E} = \begin{cases} \dfrac{Q}{4\pi\epsilon_0 r^2}\,\hat{r} & z \geq R \\ 0 & z < R \end{cases}$$

Gauss law is wrong in dynamic case!
But Gauss law derived from Columb's law

what if charge oscillate?